

DAY-4

Tasks

1.

Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Steps performed

1. Shut down the computer completely.
2. Disconnect the power cable from the PC.
3. Press the power button for a few seconds to discharge remaining power.
4. Open the computer case.
5. Located the RAM slots on the motherboard near the CPU.
6. Opened the locking clips on both sides of the RAM slot.
7. Carefully removed the RAM module by holding it from its edges.
8. Checked the RAM module and slot for dust, damage, or bent parts.
9. Aligned the notch on the RAM module with the notch in the slot.
10. Pressed the RAM firmly into the slot until the locking clips clicked into place.
11. Closed the case and reconnected the power cable.
12. Started the computer and checked whether it booted normally.

Observation

The RAM module was successfully reseated. No physical damage was observed. The computer booted normally after reinstalling the RAM.

If the RAM is not inserted correctly, the PC may show no display or produce POST beep codes.

2.

Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

Step	What to check	Expected result
1. Identify the symptom	Check for no display, flickering, artifacts, crashes, or poor performance	Clearly identify the problem
2. Check display cable	Make sure HDMI/DisplayPort cable is firmly connected to the GPU	Stable display
3. Check GPU power	Make sure PCIe power connectors are properly connected	GPU receives power
4. Reseat the GPU	Turn off PC, remove GPU, and reinstall it firmly in the PCIe slot	GPU is detected properly
5. Check drivers	Reinstall/update the graphics driver	Driver loads without errors
6. Check temperatures	Check whether the GPU is overheating during use	Temperature remains within normal operating range
7. Test another display	Try another monitor or cable	Helps eliminate monitor/cable problems
8. Confirm the fix	Run a game or graphics test and observe the system	No crashes, artifacts, or display problems

3.

Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Before disconnecting

1. Shut down the test PC.
2. Disconnect the power.
3. Open the computer case.
4. Locate the SATA SSD/HDD.
5. Disconnect the SATA data cable or power cable.
6. Start the computer.

Observation before reconnecting

Because the storage drive is disconnected, the BIOS may show that the drive is missing.

The PC may display an error such as:

- "No boot device found"
- "No bootable device"
- "Operating System not found"

The exact message depends on the motherboard and operating system.

In BIOS/UEFI, the disconnected drive should also be absent from the storage/boot-device list.

Reconnecting the drive

1. Shut down the computer again.
2. Reconnect the SATA data and power cables firmly.
3. Start the computer.
4. Enter BIOS/UEFI if necessary.
5. Check the storage/boot-device list.

Observation after reconnecting

The SSD/HDD should appear again in BIOS/UEFI. If it contains the operating system and the boot order is correct, the computer should boot normally.

For an NVMe drive, the process is similar, but the drive is installed directly into an M.2 slot rather than using SATA cables.

4.

Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

I would check the RAM first.

Reason:

Faulty or poorly seated RAM can cause:

- Random restarts
- Freezing
- Blue screens
- Failure to boot
- Application crashes

RAM is also relatively easy to inspect and reseat, making it a sensible first hardware check among RAM, GPU, and storage.

RAM diagnostic steps

1. Shut down the PC and disconnect the power.
2. Open the case.
3. Remove the RAM modules carefully.
4. Inspect the modules and slots for dust or physical damage.
5. Reseat the RAM firmly.
6. If there are multiple RAM sticks, test them one at a time.
7. Try different RAM slots if necessary.
8. Boot the PC and check whether the random restarts continue.
9. Run a memory test if available.
10. If one RAM module consistently causes problems while another works normally, the faulty module is likely responsible.

If RAM is not the problem

Next I would check the GPU, especially if restarts happen during gaming or heavy graphics use. After that, I would check the storage drive, particularly if there are boot errors, freezing, or file corruption.

Order: RAM → GPU → Storage.